THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 2

Chapter 12: **Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The field lines around a straight wire are 1

(a) parallel lines **(b)** radial lines

(c) concentric circles **(d)** spirals only

2. Electromagnetic induction was discovered by 1

(a) Ohm **(b)** Oersted **(c)** Faraday **(d)** Newton

3. A freely suspended bar magnet points 1

(a) north-south **(b)** east-west **(c)** in any direction **(d)** up-down

4. The rule to find the direction of the field due to a wire is 1

(a) right-hand thumb rule **(b)** Ohm's law

(c) Joule's law **(d)** left-hand rule

5. Which device works on electromagnetic induction? 1

(a) Bulb **(b)** Heater **(c)** Cell **(d)** Generator

6. The instrument used to find the direction of a magnetic field is a 1

(a) thermometer **(b)** voltmeter **(c)** hygrometer **(d)** compass

7. A coil of many turns of insulated wire is called 1

(a) a resistor **(b)** a cell **(c)** a fuse **(d)** a solenoid

8. The current produced by electromagnetic induction is called 1

(a) leakage current **(b)** induced current

(c) direct current only **(d)** static current

9. Iron filings arrange themselves along 1

(a) magnetic field lines **(b)** straight lines only

(c) random directions **(d)** electric lines

10. The field inside a solenoid is 1

(a) uniform **(b)** random

(c) zero **(d)** weak at the centre

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is a solenoid? 2

12. What is a generator used for? 2

13. State the direction of field lines outside a magnet. 2

14. What is an electromagnet? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. How can you show the magnetic field around a bar magnet using iron filings? 3

16. What is the principle of an electric generator? 3

17. What is the voltage of the household supply in India? Why are appliances connected in parallel? 3

OR

Describe Oersted's experiment briefly. 3

18. Distinguish between AC and DC. 3

OR

Name two safety devices used in a domestic circuit. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the magnetic field due to a current-carrying straight conductor. 5

OR

Describe a domestic electric circuit. Name the wires and the safety devices. 5

20. What is a solenoid? Describe the field pattern and state how an electromagnet is made. 5

OR

What are short circuit and overloading? How are they avoided? 5

END OF QUESTION PAPER — Total: 40 Marks

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